

Resting-State EEG Markers of Alzheimer's Disease: Spectral Power and Signal Complexity

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Abstract:

Alzheimer's disease is associated with slowing of resting-state electroencephalographic (EEG) activity, but it is unclear whether signal complexity provides additional information. The current study used R to analyze eyes-closed EEG from OpenNeuro ds004504, which includes 36 participants with Alzheimer's, 29 healthy individuals, and 23 with frontotemporal dementia. 60 seconds of cleaned EEG sampled at 250 Hz were examined from four electrodes; moreover, primary analyses used the posterior Pz and O1 signals. Furthermore, Welch spectral analysis quantified relative delta, theta, alpha, and beta power within 2-45 Hz, while permutation entropy was used to measure temporal complexity. Welch's t-test compared Alzheimer's disease and healthy controls, Pearson correlations assessed associations with MMSE scores, and sensitivity analyses tested if findings depended on electrode, frequency-range, or entropy. Comparing the groups, the Alzheimer's disease group showed higher relative delta power ($p = .005$) and theta power ($p = .002$), and lower alpha power ($p < .001$) than the controls. Beta power ($p = .102$) and permutation entropy at the predefined settings ($p = .100$) did not differ significantly. Alpha power correlated with MMSE scores across Alzheimer's and controls ($r = .421$, $p = .00048$), but did not within the Alzheimer's disease group alone ($r = .107$, $p = .534$). Alpha differences were consistent across all posterior-electrode sensitivity settings, whereas permutation-entropy results changed direction across time-delay settings. Overall, these findings indicate that EEG spectral slowing, particularly reduced alpha power, was a more accurate marker for Alzheimer's disease than permutation entropy in this dataset.

Background Information:

- Alzheimer's disease is a progressive neurodegenerative disorder and the leading cause of dementia in older adults, characterized by worsening memory, reasoning, and other cognitive abilities that can interfere with everyday activities (Safiri et al., 2024).
- EEG is a noninvasive method that records brain activity with millisecond-level temporal resolution (Ghassemkhani et al., 2025).
- Alzheimer's is commonly associated with EEG slowing, reduced alpha activity, and increased lower frequency delta and theta activity (Schwerin et al., 2025).
- Previous studies have analyzed the OpenNeuro ds004504 dataset using spectral and nonlinear EEG measurements.

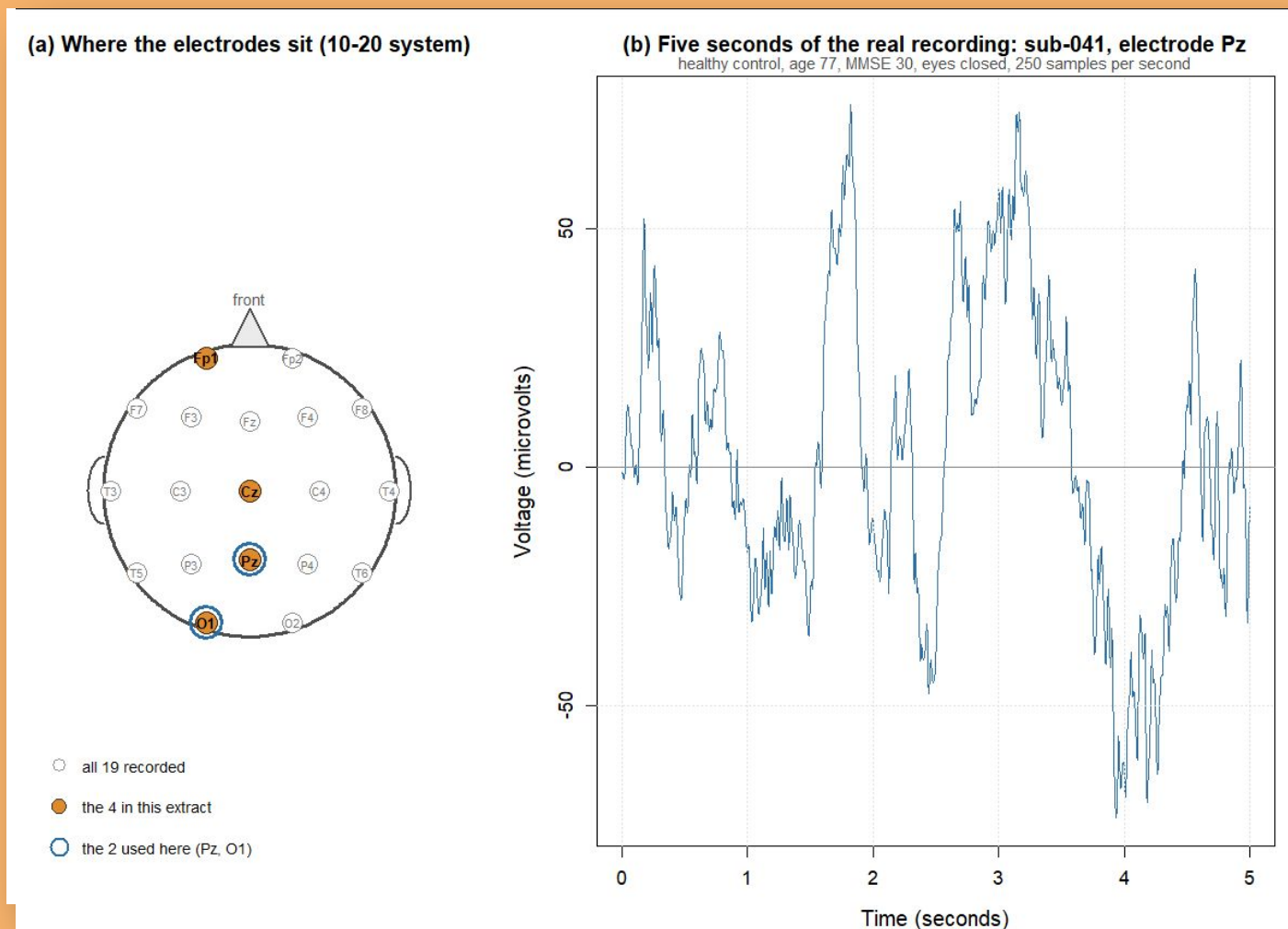


Figure 1:
a) International 10–20 electrode layout showing the four retained electrodes and the posterior analysis sites, Pz and O1.
(b) Representative five-second eyes-closed EEG recording from Pz in a healthy control. (OpenNeuro dataset ds004504)

Method/Materials:

- OpenNeuro ds004504 Dataset**
 - Resting-state, eyes-closed EEG
 - 88 participants: 36 Alzheimer's disease, 23 frontotemporal dementia, and 29 healthy controls
- EEG Recordings**
 - 60 Second per participant; Sampling rate: 250 Hz
 - Electrodes retained: Fp1, Cz, Pz, and O1. Pz and O1 averaged for primary analysis
- Spectral Analysis**
 - Welch's method to estimate power spectral density
 - Relative delta, theta, alpha, and beta power calculated within 2-45 Hz
- Complexity Analysis**
 - Permutation entropy calculated using six-point patterns spaced 20 ms apart
- Statistical Analysis**
 - Welch's t-tests for group comparisons; Cohen's d for effect size
 - Pearson correlations between EEG measures and MMSE scores
 - Sensitivity analyses tested different electrodes, frequency ranges, and entropy parameters
- R and RStudio used for analysis and figure creation

Results:

- Alzheimer's patients showed higher delta ($p = .0054$) and theta power ($p = .0021$)
- Alpha power was lower in Alzheimer's disease ($p = .00053$, $d = -1.01$)
- Beta power was not significantly different ($p = .102$)
- Permutation entropy did not differ significantly between groups ($p = .100$)
- Alpha power was not correlated with MMSE within the Alzheimer's group ($r = .107$, $p = .534$)

Relative Band Power = Power within the frequency band divided by total power from 2-45 Hz

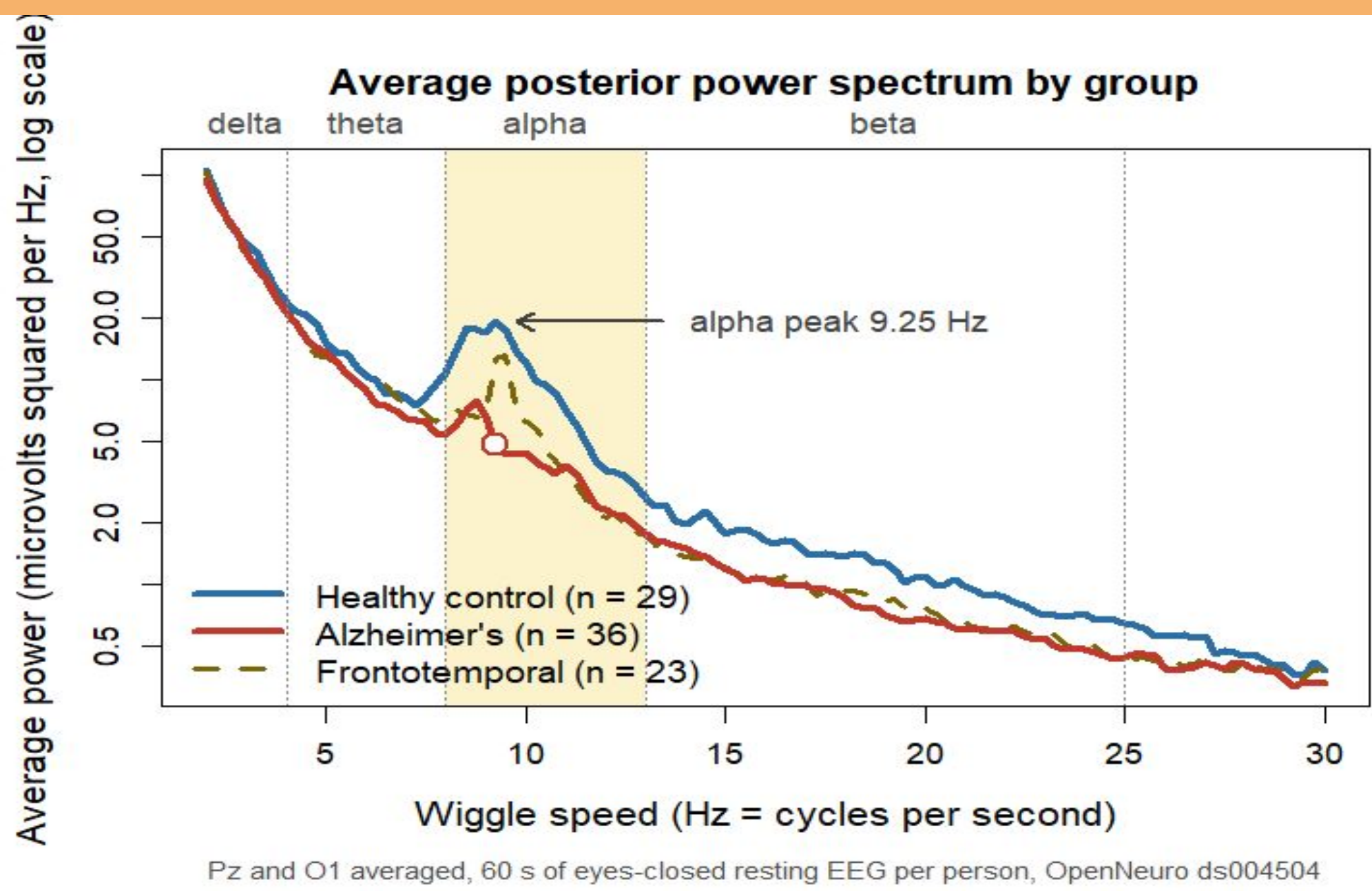


Figure 2: Average posterior power spectra by group. Healthy controls showed a stronger alpha peak, while Alzheimer's participants showed greater low-frequency power.

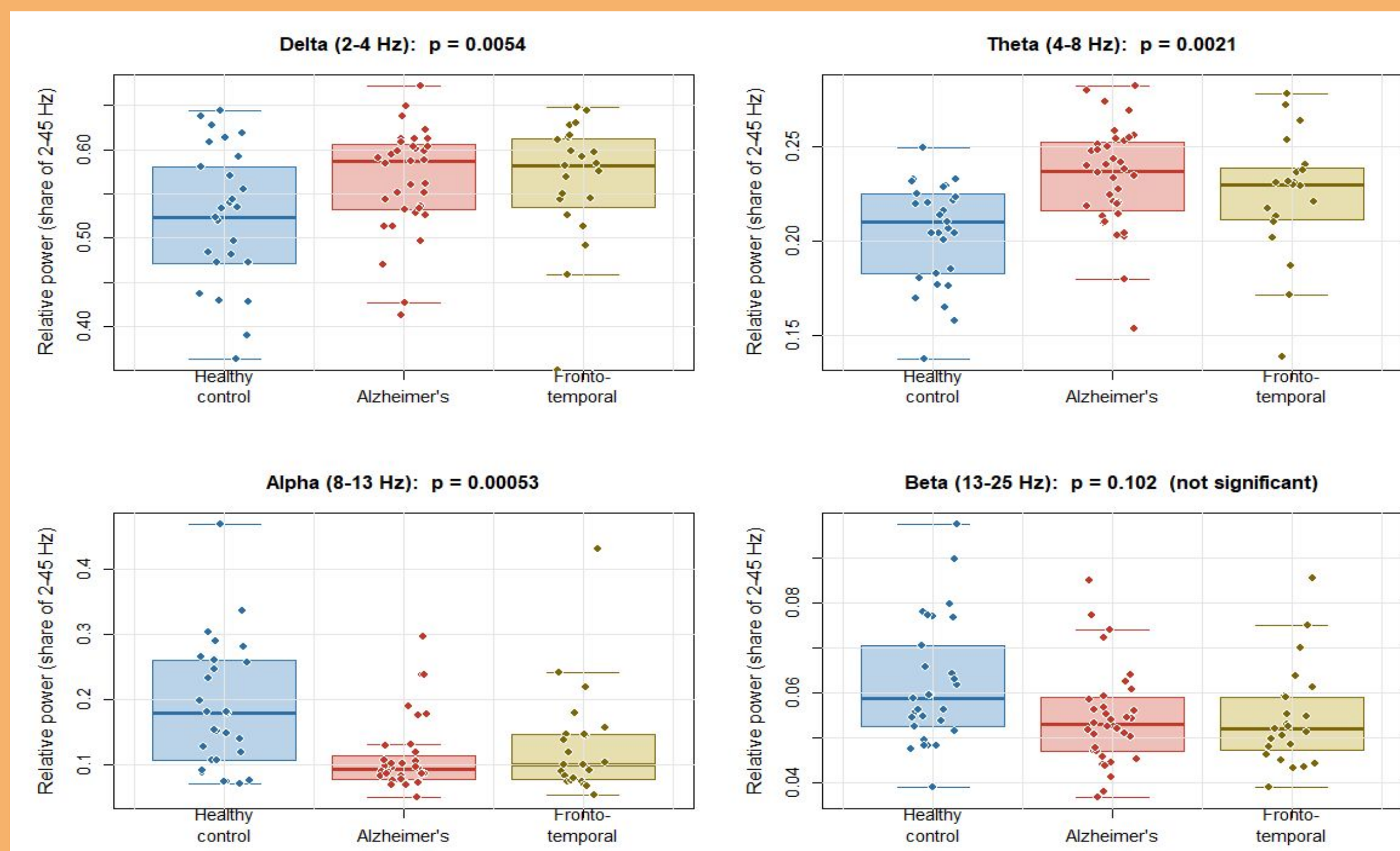


Figure 3: Relative band power by group. Alzheimer's participants had higher delta and theta power, lower alpha power, and no significant beta difference.

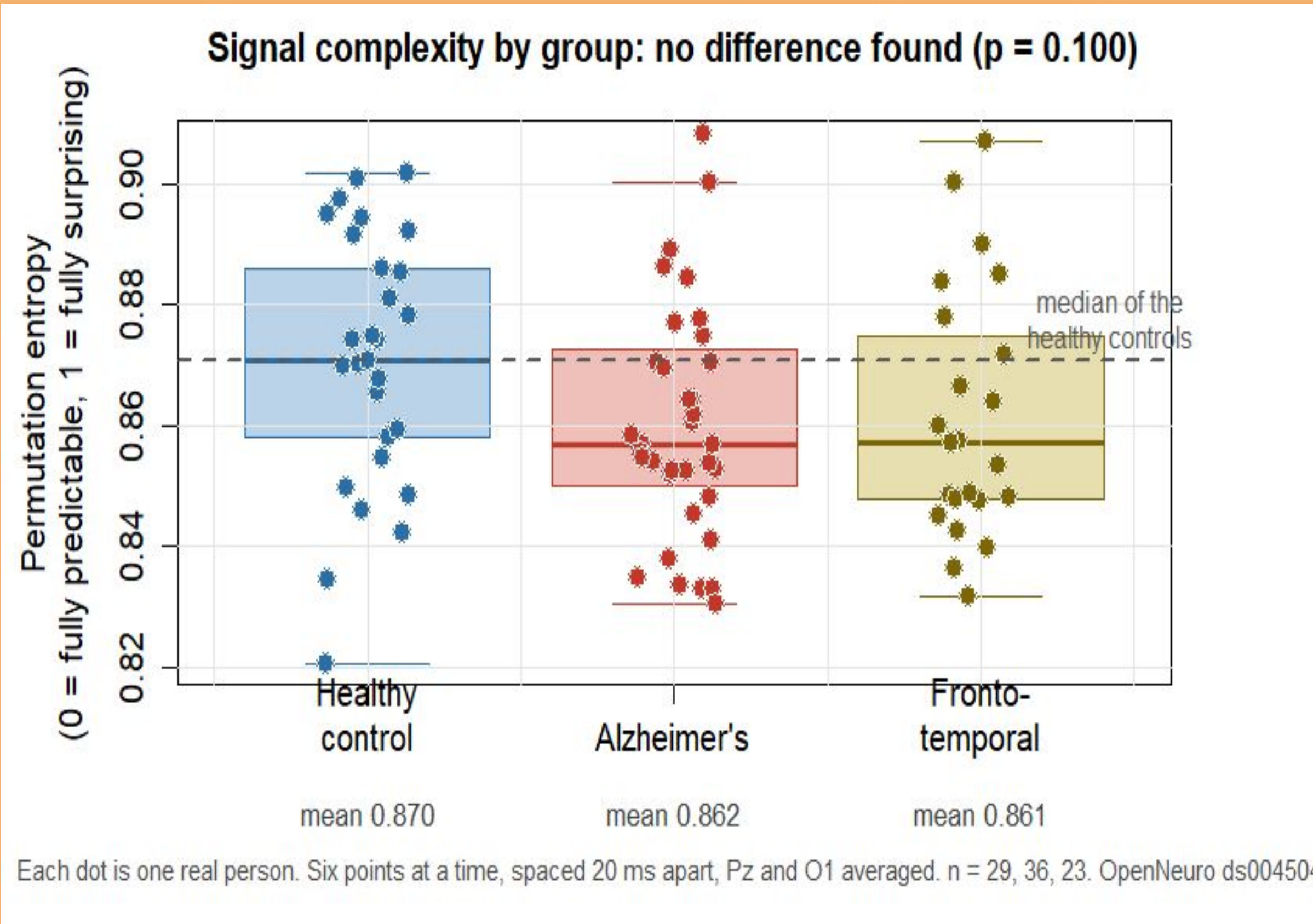


Figure 4: Permutation entropy by group. Alzheimer's and healthy-control values overlapped substantially, with no significant difference. ($p = .100$)

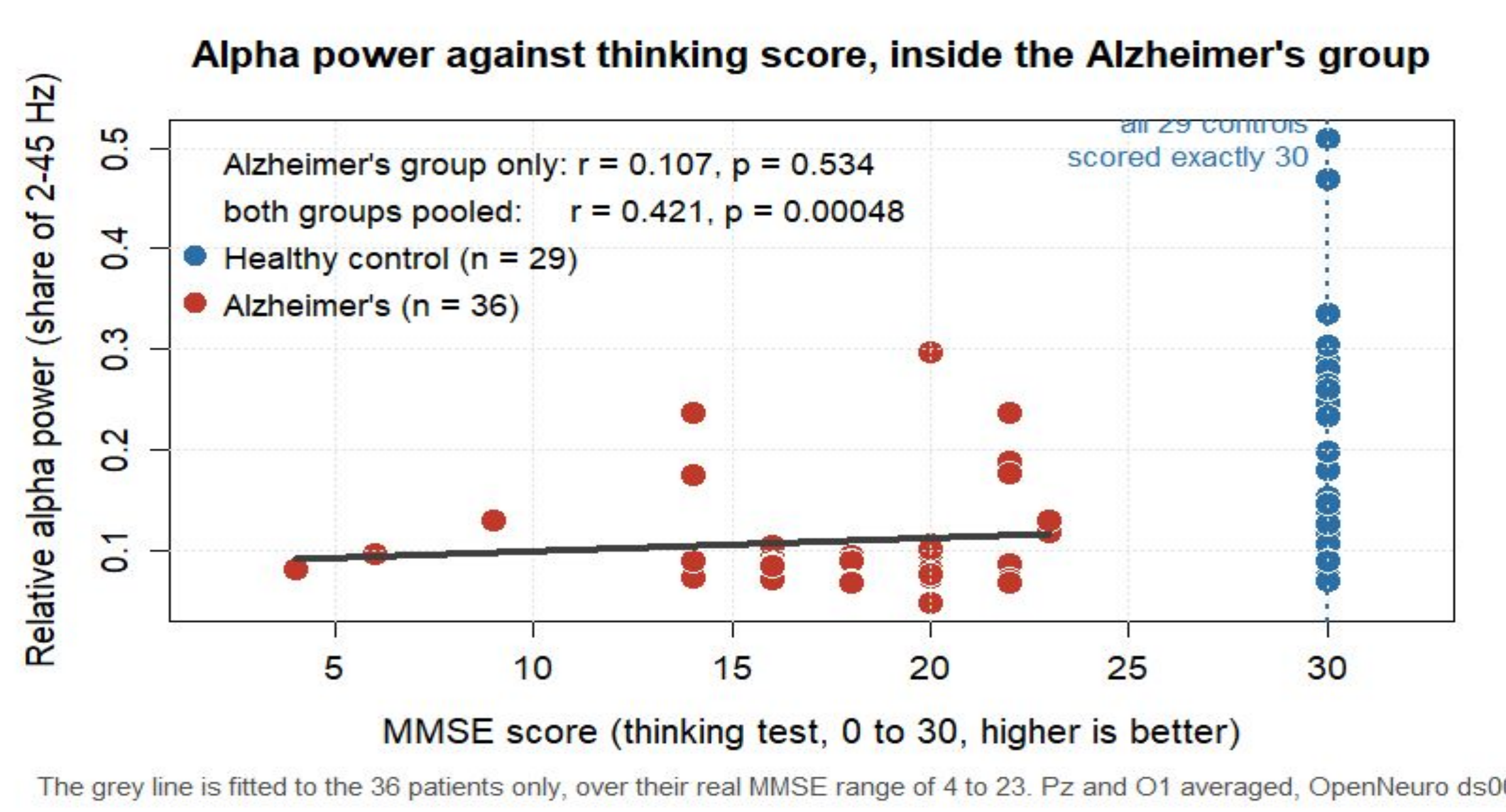


Figure 5: Alpha power versus MMSE. Alpha was not significantly associated with MMSE within the Alzheimer's group ($r = .107$, $p = .534$)

Discussion/Conclusion:

- Alzheimer's disease was associated with EEG slowing: higher delta and theta power and lower alpha power
- Alpha power remained consistent as it produced a clear difference across multiple analysis settings.
- Permutation Entropy did not significantly separate groups: altering tau and the EEG at different time scales.
- The large overlap in entropy highlights that EEG complexity varies considerably between individuals.
- Alpha power did not correlate with MMSE within the Alzheimer's group.

Limitations and Future Research

- Analysis consisted of only four electrodes (primary Pz and O1)
- Used a single dataset; Permutation entropy depends strongly on parameter selection
- Future: Use longitudinal data to study disease progression; train a formal diagnostic prediction model (combining spectral and complexity features)

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